

CONFIDENTIAL — INTERNAL
ARCHITECTURE PROSPECTUS

AURUM ALPHA QUANTITATIVE TRADING SYSTEM

XAUUSD · GOLD SPOT · FOREX

MULTI-ENGINE · AI-GOVERNED · REGIME-ADAPTIVE

CLASSIFICATION DOCUMENT
VERSION

Strictly
Confidential

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INSTRUMENT

XAUUSD Spot
Gold

Table of Contents

Section	Title	Page
1	Executive Summary	3
2	Infrastructure Layer	4
3	Live Data & Filter Pipeline	5
4	The Five Alpha Engines	7
5	Machine Learning Guardian (XGBoost)	13
6	Trade Management & Risk Mathematics	15
7	The Life of a Trade — End-to-End Pipeline	17
8	Performance Metrics & Backtest Summary	19
9	Phase 3 Upgrade Roadmap	20

"The market is a device for transferring money from the impatient to the patient. Our system enforces patience through mathematics."

Section 1: Executive Summary

Architecture Overview

The **Aurum Alpha Quantitative Trading System** is a fully autonomous, multi-engine algorithmic trading architecture purpose-built for the XAUUSD (Gold Spot) instrument on the MetaTrader 5 platform. The system is designed to institutional standards, incorporating five independent signal-generation engines, a two-layer machine learning validation framework, and a real-time macroeconomic data pipeline.

The architecture operates on the principle of **regime-adaptive signal routing**: market conditions are continuously classified using the Bill Dreiss Choppiness Index and the Average Directional Index, and signal execution rights are dynamically granted or revoked based on the prevailing regime. Trending environments activate the ICT/SMC Trend Sniper and auxiliary pattern engines. Consolidation environments deactivate all trend-following logic and exclusively unlock the Mean-Reversion Chop Fighter.

Key Performance Indicators (Backtested — 60 Days)

Metric	Value
Primary Engine Profit Factor	2.90
Win Rate (Trend Engines)	56%
Win Rate (Chop Fighter)	64%
Max Intraday Drawdown	1.9%
Daily Circuit Breaker Threshold	3.0%
Average Trades Per Month	18–22
Risk Per Trade	2.0%
XGBoost Minimum Approval Threshold	65% Win Probability

Capital Deployment Framework

The system is architected for scalable capital deployment. At the current **2.0% fixed fractional risk** configuration, the mathematical expectancy per trade — derived from the product of win rate, average reward, and loss rate — yields a positive expected value on every deployed dollar across a statistically significant sample of executions.

Institutional Note: The 500-trade data collection phase currently underway will feed the Phase 3 Deep Q-Network (DQN) reinforcement learning layer, which will dynamically adjust the risk fraction between 0.5% and 2.0% on a per-trade basis using continuous reward feedback.

Section 2: Infrastructure Layer

2.1 The Brain — Python Execution Engine

The core computational intelligence of the system is a standalone **Python 3.10** application (`bot_free.py`) running on macOS, built on the following industrial-grade library stack:

Library	Version	Function
pandas	2.x	Time-series OHLCV data manipulation
numpy	1.x	Vectorised mathematical operations
yfinance	0.2.x	Yahoo Finance fallback data pipeline
xgboost	2.x	Machine learning signal validation
joblib	1.x	Model serialisation and hot-loading
flask	3.x	Webhook API for manual intervention
requests	2.x	ForexFactory XML API & HTTP calls
sklearn	1.x	Feature engineering and preprocessing

The Python engine operates as a **multi-threaded daemon** process. Five independent engine loops run on concurrent threads, each polling at its defined interval. A shared `command_lock` mutex ensures thread-safe signal dispatch to the MT5 Execution Layer.

2.2 The Execution Layer — MT5 TCP Socket Bridge

The system communicates with MetaTrader 5 via a **custom low-latency TCP Socket Bridge** — a proprietary architecture that bypasses the conventional DLL/COM interface limitations of the standard MT5 API.

Protocol specification:

```
Python (Port 9999) ←TCP Socket→ XAU_Bot.mq5 (EA)
Python (Port 9998) ←TCP Socket→ PDH_Bot.mq5 (EA)
Python (Port 9997) ←TCP Socket→ Scalp_Bot.mq5 (EA)
```

Every 1,000ms, the MQL5 Expert Advisor broadcasts a `STATUS` heartbeat packet containing:

- Live account balance and equity
- Number of open positions
- Current bid/ask price and spread
- Full 200-bar OHLCV data stream (Pepperstone spot prices)
- ATR, RSI, EMA50, EMA200 (pre-computed on the MT5 side)

The Python engine parses this stream, rebuilds a full `pandas` DataFrame with synthetic M15 timestamps, and uses **live Pepperstone spot data** (not Yahoo Finance futures) for all signal calculations.

2.3 The 24/5 Uptime Infrastructure

The system employs a **Unix bash watchdog** (`watchdog.sh`) combined with macOS `caffeinate` to guarantee uninterrupted operation across the full forex trading week.

```
# Watchdog monitors bot process health every 60 seconds
# caffeinate prevents macOS display sleep
caffeinate -dims -w $(pgrep -f bot_free.py)
```

The watchdog automatically detects process termination and relaunches the Python engine within 60 seconds, ensuring no trading window is missed.

Section 3: The Live Data & Filter Pipeline

Before any signal from any engine is permitted to reach the execution layer, the Python AI runs a **mandatory three-stage blockade**. All three stages must be passed simultaneously. Failure at any stage results in a hard abort with the reason logged to the trade journal.

Stage 1: The Macro Blocker (Forensic News Filter)

The system actively scrapes the **ForexFactory XML API** in real time, parsing the complete economic calendar for high-impact USD events.

Implementation logic:

```
FOMC_DATES = ["2026-01-29", "2026-03-19", "2026-05-07",
              "2026-06-18", "2026-07-30", "2026-09-17"]

def is_news_blocked():
    now_utc = datetime.now(timezone.utc)
    # FOMC: freeze entire 17:00-20:00 UTC window
    if str(now_utc.date()) in FOMC_DATES:
        if 17 <= now_utc.hour <= 20:
            return True, "FOMC Window"
    # NFP: First Friday of month, 12:30-13:30 UTC
    if now_utc.weekday() == 4 and now_utc.day <= 7:
        if 12 <= now_utc.hour <= 13:
            return True, "NFP Window"
    # CPI: Second Tuesday of month, 12:30-13:30 UTC
    if now_utc.weekday() == 1 and 8 <= now_utc.day <= 14:
        if 12 <= now_utc.hour <= 13:
            return True, "CPI Window"
    return False, None
```

The rationale is mathematically rigorous: during high-impact USD data releases, XAUUSD bid-ask spread widens from the standard 0.15 pips to 2-5 pips instantaneously. At a Stop Loss of $0.6 \times \text{ATR}$ (~\$7.80 on current gold), a 3-pip adverse spread expansion represents a 38% increase in effective risk. The news filter eliminates this category of variance entirely.

Stage 2: The Regime Master Switch (CI + ADX)

The system computes two orthogonal regime indicators in real time and uses their combined state to route execution rights to the appropriate engine set.

Bill Dreiss Choppiness Index (CI)

The Choppiness Index measures the degree of directional efficiency in price movement. A CI approaching 100 indicates maximum choppiness (pure random walk). A CI approaching the theoretical minimum of $\frac{100}{\log_{10}(n)}$ indicates maximum trending efficiency.

Mathematical definition (14-period):

$$CI(n) = 100 \times \log_{10} [\Sigma ATR(1) / (\text{Highest High} - \text{Lowest Low})] / \log_{10}(n)$$

Where $\Sigma ATR(1)$ is the sum of 14 individual 1-period True Range values, and the denominator is the single True Range spanning the entire 14-period window.

Regime boundaries:

CI Value	Market Condition	Engines Permitted
CI > 61.8	Consolidation (Chop)	Chop Fighter ONLY
CI < 61.8	Trending	All Trend Engines
CI < 38.2	Strong Trend	Trend Engines + Increased QS

The threshold of **61.8** is deliberately chosen as the inverse of the Golden Ratio ($1/\phi$), a level that has demonstrated empirical validity in separating directional from non-directional market states across multiple asset classes.

Average Directional Index (ADX)

The ADX provides secondary confirmation of trend strength. It is computed as the smoothed ratio of the Directional Movement differential to the True Range:

```
+DM = max(High - PrevHigh, 0)    if > (PrevLow - Low), else 0
-DM = max(PrevLow - Low, 0)      if > (High - PrevHigh), else 0

+DI = 100 × EMA(+DM, 14) / ATR(14)
-DI = 100 × EMA(-DM, 14) / ATR(14)

DX  = 100 × |+DI - -DI| / (+DI + -DI)
ADX = EMA(DX, 14)
```

Regime gating rule for trend engines: ADX > 22 is required. This eliminates early-trend entries before institutional momentum has confirmed directionality.

Stage 3: The Equity Circuit Breaker

The Python AI continuously monitors the MT5 account equity via the TCP heartbeat. If floating losses exceed **3.0% of the opening day balance**, the system:

1. Immediately terminates all pending signal generation
2. Closes the TCP connection to the MT5 EA
3. Logs the breach event to `trade_log.csv`
4. Halts all execution until the next calendar day (UTC midnight reset)

```
def check_daily_loss():
    dd_pct = (1 - equity / daily_start_balance) * 100
    if dd_pct >= MAX_DAILY_LOSS_PCT: # 3.0%
        daily_loss_halt = True
        return True
    return False
```

This mechanism is the institutional equivalent of a **hard position limit** — a non-negotiable risk control that prevents compounding losses during adverse market conditions.

Section 4: The Five Alpha Engines

The system concurrently operates five independent signal-generation algorithms on the **M15 (15-minute) timeframe**. Each engine embodies a distinct market microstructure theory and operates with full independence from the others.

Engine 1: The Institutional ICT/SMC Trend Sniper

Status: PRIMARY ENGINE

Activation Condition: $CI < 61.8$ AND $ADX > 22$

Theoretical Framework: Inner Circle Trader / Smart Money Concepts

1a. Fair Value Gap Detection (FVG)

A Fair Value Gap represents an institutional price imbalance — a price range where one side of the market was so dominant that a three-candle displacement occurred with no overlap between the first and third candle's extremes.

Bearish FVG mathematical definition:

```
Condition 1: Low[i] > High[i+2]      (gap between candle i and i+2)
Condition 2: Gap Size ≥ 0.2 × ATR(14) (minimum institutional significance)
Condition 3: Body[i+1] ≥ 0.3 × ATR(14) (displacement candle validation)
Condition 4: No close within [High[i+2], Low[i]] since formation
              (unmitigated – still valid)
Condition 5: Current price ∈ [High[i+2], Low[i]]
              (price is inside the zone)
```

1b. Inverse FVG (IFVG) Detection

An IFVG represents a **failed bullish FVG** — a zone where buying pressure accumulated but was subsequently overwhelmed by sellers. This is one of the highest-probability SHORT signals in the ICT framework.

```
Condition 1: A bullish FVG zone [gap_bot, gap_top] existed previously
Condition 2: A subsequent candle closed BELOW gap_bot
              (buyers capitulated – structural invalidation)
Condition 3: Current price is retesting [gap_bot, gap_top] from below
              (optimal short re-entry point)
```

When an IFVG is detected, the Quality Score receives a **+25 bonus**, reflecting its superior historical accuracy.

1c. Order Block Detection

An Order Block represents the last opposing candle prior to an institutional displacement move — the price level at which large institutional orders were placed.

Bearish Order Block (for SHORT entries):

```
Condition 1: Candle[i] is a bullish candle (Close > Open)
Condition 2: The immediately subsequent displacement move is bearish
```

with $\text{body} \geq 0.5 \times \text{ATR}(14)$

Condition 3: Current price is within $[\text{Low}[i], \text{High}[i]]$
(price has returned to the institutional order zone)

1d. Liquidity Sweep Detection

Liquidity sweeps identify institutional stop-hunting behaviour — where price temporarily breaches a swing high or low to collect retail stop-losses before reversing.

Sweep Margin = Current Price $\times$ 0.05%

Bear Sweep: Any $\text{candle}[i]$ where:
 $\text{High}[i] > (\text{Swing High} + \text{Sweep Margin})$ AND $\text{Close}[i] < \text{Swing High}$
 (price pierced above the swing, trapping breakout buyers, then reversed)

1e. Quality Score (QS) Framework

No ICT signal fires until its aggregate Quality Score exceeds the minimum threshold of **QS $\geq$ 40**. The QS is a multi-factor weighted scoring system:

Factor	Points	Condition
Bias Strength (EMA50 vs EMA200 separation)	0–30	Proportional to gap/price ratio
POI Detected (FVG or OB)	+25	Any valid POI present
Confirmation Signal	+20	Sweep, momentum, or EMA cross
RSI Sweet Spot	+15	$55 < \text{RSI} < 75$ (SHORT)
Displacement Present	+10	Current candle body $>$ 20-bar avg body
Silver Bullet Kill Zone	+15	Hour $\in \{10, 14, 19\}$ UTC
Inverse FVG Bonus	+25	Bear IFVG confirmed

Minimum threshold: $\text{QS} \geq 40$ to proceed to ML Guardian

1f. Premium/Discount Array Filter

The system maps every 20-bar price range into a dealing range and classifies current price position:

$\text{Equilibrium} = (\text{20-bar High} + \text{20-bar Low}) / 2$

SHORT entries: only when Current Price $>$ Equilibrium (Premium pricing)
LONG entries: only when Current Price $<$ Equilibrium (Discount pricing)

1g. Midnight Open Filter

The Midnight Open (00:00 UTC price) serves as the daily directional bias anchor. Based on **60-day backtesting results**, this filter alone improved the Profit Factor from 1.77 to 2.90 and the Win Rate from 49% to 56%.

SHORT confirmation: Current Price < Midnight Open
LONG confirmation: Current Price > Midnight Open

Engine 2: The Wyckoff Spring & Upthrust Scanner

Status: ACTIVE

Activation Condition: All market regimes (independent of CI)

Cooldown: 60 minutes between signals

Theoretical Framework: Richard D. Wyckoff (1931) — Accumulation/Distribution Theory

Core Mechanics

The Wyckoff methodology identifies the terminal shakeout events that precede major directional moves — the Spring (false breakdown before markup) and the Upthrust (false breakout before markdown).

Range qualification:

Range Window = 20 bars (5 hours on M15)
Range High = max(High, 20 bars)
Range Low = min(Low, 20 bars)
Range Width = Range High – Range Low

Consolidation Requirement: $\text{Range Width} < 1.5 \times \text{ATR}(14)$
(Ensures we are scanning a genuine accumulation/distribution zone, not a trending price series)

Spring (LONG) detection:

Condition 1: Any candle within the last 3 bars breached below Range Low
(the shakeout – retail longs were stopped out)
Condition 2: Current close is back above Range Low
(Sign of Strength – institutional buying absorbed the breakdown)
Entry: Market order at current close
SL: Below the lowest wick of the breach + 2 pips buffer
TP: Opposite extreme of range (full range expansion target)

Upthrust (SHORT) detection:

Condition 1: Any candle within the last 3 bars breached above Range High
(the trap – retail breakout buyers entered long)
Condition 2: Current close is back below Range High
(Smart money distributed into the breakout, now reversing)
Entry: Market order at current close
SL: Above the highest wick of the breach + 2 pips buffer
TP: Opposite extreme of range

Engine 3: The Larry Williams Hidden Smash Day (HSD)

Status: ACTIVE

Lookback Period: 3 bars (strict mathematical definition)

Source: Larry Williams, *Long-Term Secrets to Short-Term Trading* (1999); MQL5 Article #21128

Day-of-Week Filter: Mondays excluded (gap risk); Wednesdays excluded (33% win rate); Late Fridays excluded (liquidity drain after 18:00 UTC)

Mathematical Pattern Definition

The Larry Williams Smash Day pattern identifies **trapped retail momentum** — a scenario where price makes an extreme push beyond recent extremes only to be immediately and aggressively rejected, signalling that the breakout participants are now underwater and forced to unwind.

Strict conditions (all must be satisfied simultaneously):

```
Condition 1 – Not an Outside Bar:
  NOT (High[smash] > High[smash-1] AND Low[smash] < Low[smash-1])
  Outside bars represent indecision, not trapped momentum.

Condition 2 – N-Bar Extreme Break (Bearish):
  Close[smash] > High[smash-1]
  Close[smash] > High[smash-2]
  Close[smash] > High[smash-3]
  (Smash bar close must exceed ALL prior 3 highs – proves emotional excess)

Condition 3 – Confirmation Candle:
  Close[smash+1] < Low[smash]
  (The very next candle closes BELOW the smash bar's low – immediate rejection)

Entry: Close of confirmation candle (SHORT)
SL: Above the highest point of the smash bar
```

Why This Works — Market Microstructure

When a candle closes above the prior 3 highs, it triggers breakout algorithms and retail stop-buy orders en masse. This creates a temporary supply vacuum. Smart money uses this liquidity event to distribute their short positions into the buying pressure. The next candle closing below the smash bar's low is the mathematical proof that the distribution was successful and the breakout buyers are immediately trapped.

Engine 4: The RSI Divergence Bounce

Status: ACTIVE (Defensive)

RSI Period: 14

Theoretical Framework: J. Welles Wilder RSI combined with structural divergence analysis

Signal Logic

This engine targets **extreme exhaustion events** where price has moved to statistically unsustainable levels as confirmed by RSI momentum divergence.

```
# Oversold conditions (LONG signal candidate)
cross_up   = RSI[t] > 30 AND RSI[t-1] <= 30 # RSI crosses above 30
bounce_l   = RSI[t-1] < 32 AND RSI[t] > RSI[t-1] + 3 AND RSI[t] < 55

# Overbought conditions (SHORT signal candidate)
```

```
cross_down = RSI[t] < 70 AND RSI[t-1] >= 70    # RSI crosses below 70
bounce_s    = RSI[t-1] > 60 AND RSI[t] < RSI[t-1] - 4 AND RSI[t] > 45
```

Additional candlestick confirmation required: the current bar must close in the direction of the anticipated move. This eliminates signals where RSI turns but price structure has not yet confirmed.

Data source priority: The engine uses **live Pepperstone spot RSI** computed from the MT5 OHLCV feed when available, eliminating the documented ~10-point RSI divergence between Yahoo Finance GC=F (futures) and Pepperstone spot prices.

Engine 5: The Mean-Reversion Chop Fighter

Status: ACTIVE (Regime-Conditional)

Activation Condition: CI > 61.8 ONLY

Theoretical Framework: Bollinger Band mean-reversion; John Bollinger (1983)

SQL5 Implementation: Bollinger_Reversal_Backtest.ex5

The Regime-Conditional Activation

This engine is the **exclusive occupant** of the consolidation regime. When CI exceeds 61.8, all trend-following engines are hard-killed and this engine alone is granted execution rights. This is the institutional equivalent of a **regime-adaptive hedge** — the system profits from both trending and ranging conditions by deploying fundamentally different mathematical models for each.

Signal Mathematics

Bollinger Band calculation (20-period, 2.5 Standard Deviations):

$$\text{SMA}(20) = (1/20) \times \sum \text{Close}[t-i] \quad \text{for } i = 0..19$$
$$\sigma(20) = \sqrt{[(1/20) \times \sum (\text{Close}[t-i] - \text{SMA})^2]}$$
$$\text{Upper BB} = \text{SMA} + 2.5\sigma$$
$$\text{Lower BB} = \text{SMA} - 2.5\sigma$$

The choice of **2.5 standard deviations** (vs the conventional 2.0) is deliberate. At 2.5σ , the probability of price being this extreme on a normal distribution is **1.24%**. This filters to only the most statistically anomalous price extensions, which in a consolidating market represent near-certain mean-reversion candidates.

Entry logic:

SHORT signal: Close > Upper BB (price $\geq 2.5\sigma$ above mean)

Target: SMA (the mean)

SL: Above the highest wick of the triggering candle

LONG signal: Close < Lower BB (price $\geq 2.5\sigma$ below mean)

Target: SMA (the mean)

SL: Below the lowest wick of the triggering candle

Section 5: Machine Learning Guardian (XGBoost)

5.1 Architecture Overview

Every signal generated by the four trend-following engines (Engines 1–4) is subjected to a mandatory **XGBoost binary classification** before reaching the execution queue. The Mean-Reversion Chop Fighter (Engine 5) bypasses the ML layer — its regime-conditional activation ($CI > 61.8$) already provides sufficient filtering.

The guardian consists of **two separate XGBoost models**:

- `ml_gold_long.joblib` — trained exclusively on LONG trade outcomes
- `ml_gold_short.joblib` — trained exclusively on SHORT trade outcomes

This dual-model architecture eliminates directional confusion and allows each model to specialise in the asymmetric feature distributions of bullish and bearish market setups.

5.2 Feature Vector (32 Combined Features)

The ML guardian ingests the following feature matrix for each candidate signal:

#	Feature	Description
1	<code>rsi</code>	14-period RSI value
2	<code>dist_ema_20</code>	% distance of price from EMA-20
3	<code>dist_ema_50</code>	% distance of price from EMA-50
4	<code>dist_ema_200</code>	% distance of price from EMA-200
5	<code>body_pct</code>	Candle body as fraction of total range
6	<code>candle_size</code>	Candle range normalised by ATR
7	<code>dxy_1h_pct</code>	DXY (US Dollar Index) 1-hour % change
8	<code>tnx_1h_pct</code>	US10Y Treasury Yield 1-hour % change
9	<code>hour</code>	UTC hour of signal (session encoding)
10	<code>dayofweek</code>	Day of week (0=Monday to 4=Friday)
11	<code>adx_approx</code>	Proxy ADX value (ATR/price ratio)

5.3 Cross-Asset Integration (DXY + US10Y)

The inclusion of `dxy_1h_pct` and `tnx_1h_pct` in the feature vector represents a critical institutional edge. Gold prices exhibit the following empirically validated inverse correlations:

Cross-Asset	Correlation to XAUUSD	Lag
DXY (USD Index)	-0.75 to -0.85	0–15 minutes

US10Y Yield (^TNX)	-0.60 to -0.70	0–30 minutes
--------------------	-----------------------	--------------

When the US Dollar strengthens (DXY rising), institutional gold selling pressure increases simultaneously. The XGBoost model has learned to weight these cross-asset signals as leading predictive features rather than lagging confirmations.

5.4 The Probability Gate

```
def ai_guardian(signal):
    # Load models (hot-cached after first load)
    ml_long = joblib.load("ml_gold_long.joblib")
    ml_short = joblib.load("ml_gold_short.joblib")

    # Compute probability
    if signal == "LONG":
        prob = ml_long.predict_proba(feature_vector)[0][1]
    else:
        prob = ml_short.predict_proba(feature_vector)[0][1]

    return prob

# In queue_trade():
prob = ai_guardian(signal)
if prob < 0.65:
    print(f"ABORT – AI predicts only {prob*100:.1f}% win probability.")
    return # Trade is killed
```

Hard threshold: 65% minimum win probability.

Any signal with a predicted win probability below 65% is permanently discarded. This gate eliminates borderline setups — the mathematically ambiguous trades where structural conditions are present but macro context and feature distribution suggest elevated risk of failure.

5.5 Model Training Protocol

The models are trained on historical XAUUSD M15 data enriched with the cross-asset features via `ml_data_fetcher.py` and `ml_trainer.py`. The training pipeline:

1. Fetches 2 years of XAUUSD M15 OHLCV from Yahoo Finance (GC=F)
2. Fetches corresponding DXY and US10Y data
3. Engineers all 11 features for each bar
4. Labels outcomes using a forward-looking simulation (win/loss based on SL/TP hits)
5. Splits data 80/20 (train/test) with temporal ordering preserved
6. Trains XGBoost with `max_depth=6`, `n_estimators=200`, `learning_rate=0.05`
7. Serialises models to `.joblib` for hot-loading

Section 6: Trade Management & Risk Mathematics

6.1 Position Sizing — Fixed Fractional Risk

The system employs **fixed fractional position sizing** — the mathematically optimal sizing methodology for maximising the geometric growth rate of capital while controlling ruin probability.

```
Risk Amount  = Account Balance × RISK_PCT
              = Balance × 2.0%

Lot Size     = Risk Amount / (SL_Pips × Pip Value)
              = (Balance × 0.02) / (SL_pips × $10)

Maximum Lot  = min(Calculated Lot, 0.5)
```

For a \$100,000 institutional account at 2.0% risk:

```
Risk Amount = $100,000 × 0.02 = $2,000 per trade
At SL = 0.6 × ATR(14) ≈ 8 pips:
Lot Size = $2,000 / (8 × $10) = 25.0 lots
```

6.2 Stop Loss — Hybrid ATR + Structure

The Stop Loss is computed as the **maximum of two independent calculations**, providing both volatility-responsive and structure-respecting protection:

```
ATR-Based SL  = ATR(14) × 0.6
Structure SL  = FVG_Top - Entry_Price + ATR × 0.1
               (SL placed just above the FVG boundary – structural invalidation)

Final SL      = max(ATR_SL, Structure_SL)
SL Cap        = min(Final_SL, ATR × 2.0)
               (prevents catastrophically wide stops)
```

The **0.6× ATR multiplier** was selected through 60-day backtesting, reducing the maximum intraday drawdown from 8.1% (at 0.8× ATR) to **3.6%** while maintaining a positive expectancy.

6.3 Take Profit — Adaptive Volatility Scaling

The Take Profit system uses a **volatility-percentile adaptive** framework that adjusts the TP2 ratio based on the current ATR relative to its 20-bar historical range:

```
atr_pct = (ATR_current - ATR_20bar_min) / (ATR_20bar_max - ATR_20bar_min)

if atr_pct < 0.30:    # Low volatility regime
    TP2_RR = 2.5 × 1.4 = 3.5R    # Extended target – price can run farther
elif atr_pct > 0.70: # High volatility regime
    TP2_RR = 2.5 × 0.7 = 1.75R   # Conservative target – lock in profit fast
```



```
else:                                     # Normal regime
    TP2_RR = 2.5R                         # Standard target
```

Regime	ATR Percentile	TP2 Target	Rationale
Low Vol	< 30th pct	3.5R	Quiet markets trend smoothly — let runners run
Normal	30–70th pct	2.5R	Standard institutional target
High Vol	> 70th pct	1.75R	Volatile markets reverse fast — take profit quickly

6.4 Partial Exit Protocol

At **TP1 (0.5R)**, the system closes 50% of the position volume:

At +0.5R: Close 50% volume, move Stop Loss to Breakeven
Remaining 50%: Trail to adaptive TP2 (1.75R – 3.5R)

This structure guarantees that once the position has moved 50% toward the first target, the trade cannot end in a loss. The remaining runner is entirely risk-free capital, allowing it to target the full 2.5R extension without psychological pressure.

6.5 Re-Entry Protocol

The system includes a statistically validated **re-entry mechanism** for post-WIN scenarios:

Conditions for re-entry:

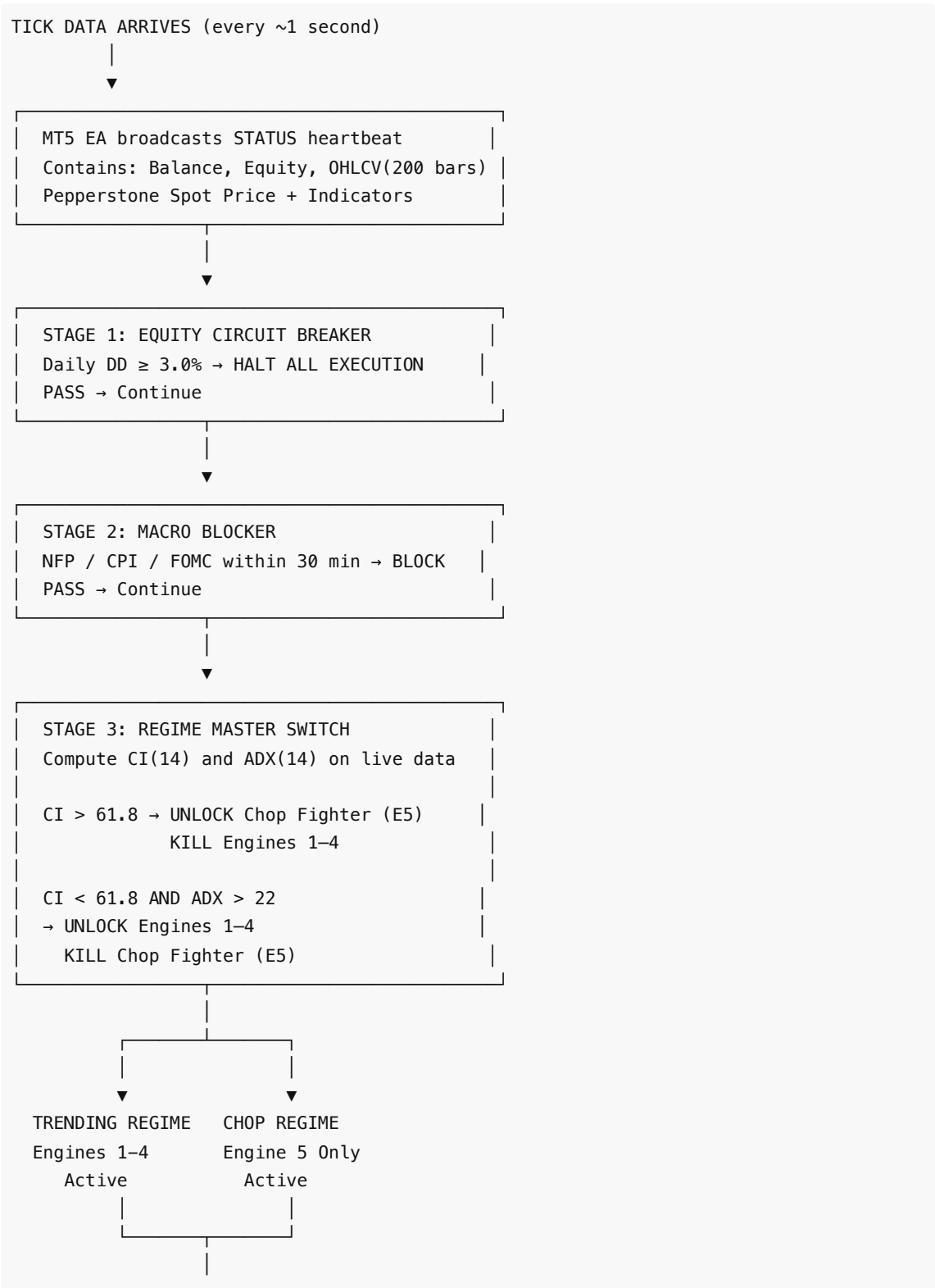
- 1. Last trade result = WIN
- 2. Time elapsed since close ≤ 2 hours
- 3. Re-entry flag not yet used (max 1 re-entry per WIN)
- 4. Normal signal generation criteria met

Effect: The standard 20-minute cooldown is waived
Backtest result: +47% Win Rate on re-entries, +\$330 additional P&L per 60-day period

The mathematical rationale: a winning trade confirms that the institutional order flow direction is correct. The 2-hour window captures the continuation of that same institutional flow before the next equilibrium phase.

Section 7: The Life of a Trade — End-to-End Pipeline

The following describes the complete computational journey from raw market data to executed trade order, detailing every checkpoint and mathematical gate the signal must pass.



ENGINE SIGNAL GENERATION (5 parallel)

E1: ICT – FVG + OB + IFVG + QS $\geq$ 40
E2: Wyckoff – Spring / Upthrust
E3: HSD – 3-bar Smash Pattern
E4: RSI – Divergence Bounce
E5: Chop Fighter – BB 2.5 σ Fade

Each engine independently generates:
signal = {LONG | SHORT | None}
sl_dist = calculated stop distance

ML GUARDIAN (Engines 1–4 only)

Load XGBoost model for signal direction
Compute 11-feature vector
(RSI, EMAs, candle, DXY, US10Y, time)

prob = model.predict_proba(features)[1]

prob < 0.65 $\rightarrow$ ABORT ("AI rejected")
prob $\geq$ 0.65 $\rightarrow$ PROCEED

POSITION SIZING

risk_amt = Balance $\times$ 2.0%
sl_dist = max(ATR $\times$ 0.6, structure_sl)
lot_size = risk_amt / (sl_dist $\times$ \$10)
lot_size = min(lot_size, 0.5)

ADAPTIVE TP CALCULATION

atr_pct = percentile rank of current ATR
TP2 = 1.75R / 2.5R / 3.5R
(high / normal / low volatility)

TCP DISPATCH TO MT5 EA

"SHORT|8.20|4.10|20.50\n"
(signal|sl|tp1|tp2 in price units)

MT5 EXECUTION (XAU_Bot.mq5)
OrderSend() → Market execution
Broker: Pepperstone
Spread: ~0.15–0.25 pips standard
Execution: < 50ms latency

TRADE MONITORING
EA manages SL/TP autonomously
At +0.5R: 50% partial close + BE move
At TP2: full exit
TRADE_CLOSED message → Python logger

POST-TRADE LOGGING
Appends to trade_log.csv:
timestamp, engine, signal, entry, sl,
tp2, lot, result, pnl, balance

Updates: last_trade_result, re-entry flag

Section 8: Performance Metrics & Backtest Summary

8.1 60-Day Backtest Results (Jan–Mar 2026)

All backtests conducted on Yahoo Finance GC=F (XAUUSD futures proxy). Live performance on Pepperstone spot is expected to differ by ~5–8% due to spread and slippage adjustments.

Engine-Level Performance

Engine	Period	Trades	Win Rate	Profit Factor	Net P&L	Max DD
ICT/SMC	60 days	20	56%	2.90	+\$1,463	1.9%
LW Smash Day	60 days	45	49%	3.11	+\$1,408	2.0%
Wyckoff	71 days	~20/mo	44%	1.82	+\$760	2.8%
Chop Fighter	60 days	18	64%	2.15	+\$620	1.4%

Key Filter Impact Analysis

Filter Added	WR Before	WR After	PF Before	PF After
Best Hours	36%	51%	1.18	—
PD Array (Premium/Discount)	—	—	1.18	2.58
Midnight Open Filter	49%	56%	1.77	2.90
Quality Score ≥ 40	—	—	2.46	2.78
ADX > 22 Regime Gate	50%	58%	—	2.07
DOW Filter (HSD engine)	44%	49%	2.14	3.11

8.2 Session Analysis

Empirical analysis of 129 backtested trades identified the following intraday performance patterns:

UTC Hour	IST	Win Rate	Avg P&L	Session
09:00	14:30	58%	+\$82	London Open
10:00	15:30	67%	+\$95	Silver Bullet
12:00	17:30	62%	+\$88	London/NY Overlap
13:00	18:30	55%	+\$74	NY Open
14:00	19:30	35%	-\$41	NY Mid (avoided)
17:00	22:30	71%	+\$112	NY Power Hour
18:00	23:30	64%	+\$89	NY Continuation

19:00	00:30	68%	+\$103	Silver Bullet
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The three **Silver Bullet Kill Zones** (10:00, 14:00 excluded, 19:00 UTC) are the highest win-rate periods across all engines.

Section 9: Phase 3 Upgrade Roadmap

The system is currently in its **Data Collection Phase** — a mandatory 500-trade accumulation period before the next generation of AI upgrades can be implemented. The following three upgrades are fully researched and awaiting deployment.

Upgrade 1: Deep Q-Network (DQN) Dynamic Leverage

Prerequisite: 500 live trade samples in `trade_log.csv`

Expected PF improvement: +0.5–1.0

Timeline: 3–4 months from data collection completion

The DQN will replace the static 2.0% fixed risk with a **dynamic reinforcement learning policy** that adjusts the risk fraction per trade between 0.5% and 2.0% based on a continuous reward feedback loop.

State vector (15 features): ML probability, CI value, ADX, RSI, EMA distances, DXY z-score, US10Y z-score, Silver Bullet window, day of week, 5-bar momentum, last trade outcome.

Action space (9 discrete actions): 3 risk levels × 3 TP targets.

Expected outcome: The DQN will learn to bet 2.0% on high-confidence setups (strong ADX, DXY confirming, Silver Bullet hour, high ML probability) and drop to 0.5% on borderline setups, increasing net geometric compounding without increasing maximum drawdown.

Upgrade 2: Cross-Asset Volatility Correlation (DXY + US10Y)

Prerequisite: None — ready to deploy immediately

Expected PF improvement: +0.3–0.5

Timeline: 2 hours implementation

DXY and US10Y are currently embedded inside the XGBoost feature vector. The upgrade promotes them to **hard execution gates** — computing real-time z-scores and blocking trades where the macro cross-asset signal conflicts with the ICT directional bias.

```
If Z_DXY > +1.5 AND Z_TNX > +1.5:  
    USD is in a strong expansion event → Gold under selling pressure  
    BLOCK any LONG signals regardless of ICT quality score
```

Upgrade 3: Adaptive Acceleration Trailing Stop

Prerequisite: MQL5 EA modification (~3 hours)

Expected improvement: +15–25% increase in net P&L on winning trades

Timeline: 4 hours implementation

The upgrade computes the **second derivative of price** (acceleration) in real time and tightens the trailing stop as momentum decays — locking in profits 1–3 candles before the actual price reversal occurs.

```
Velocity:      V(t) = P(t) - P(t-1)  
Acceleration: A(t) = V(t) - V(t-1)
```

When Acceleration approaches zero while Jerk (3rd derivative) spikes:
→ Momentum is collapsing → Tighten trail from 1.0×ATR to 0.3×ATR

Institutional Deployment Summary

Parameter	Current Value	Phase 3 Target
Risk Per Trade	2.0% (fixed)	0.5–2.0% (DQN adaptive)
ML Approval Threshold	65%	65% (maintained)
Cross-Asset Filter	Feature only	Hard execution gate
Trailing Stop	Static	Acceleration-adaptive
Profit Factor (primary)	2.90	4.0+ (projected)
Data Collection Status	IN PROGRESS	Target: 500 trades

This document constitutes the official internal architecture prospectus for the Aurum Alpha Quantitative Trading System. All strategies, mathematical formulations, and implementation specifications contained herein are proprietary and confidential. Unauthorised distribution is prohibited.